

PYTHON FOR BEGINNERS · DAY 4

While Loops

Making Python repeat things

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```
# count to 3
count = 1
while count <= 3:
    print(count)
    count += 1

# 1 2 3
```

TODAY'S GOALS

By the end of today, you can…

- Explain why we need loops
- Avoid (and stop!) infinite loops
- Use **break** and **continue**
- Write a **while** loop with a counter
- Keep asking until the answer is valid
- Upgrade our guessing game 🎮

QUICK REVIEW · DAY 3

Python can make decisions

```
score = 75
if score >= 80:
    print("Grade A")
elif score >= 50:
    print("Pass")
else:
    print("Fail")
```

if runs code only when the condition is True

elif checks another condition

else catches everything else

But each block runs **only once**…

WHY LOOPS?

Print “Hello” 5 times… or 1,000?

Without a loop 😩

```
print("Hello")
print("Hello")
print("Hello")
print("Hello")
print("Hello")
```

With a loop 😎

```
count = 1
while count <= 5:
    print("Hello")
    count += 1
```

Change 5 to 1000 — still 4 lines!

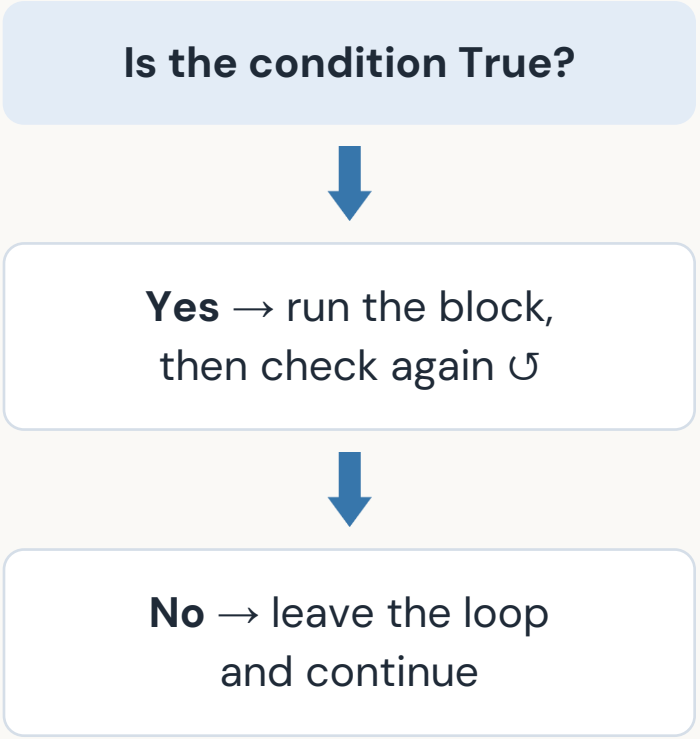
THE WHILE LOOP

Repeat while the condition is True

```
while condition:
    statement
```

Looks like **if** — but instead of running **once**, it runs **again and again**.

Same rules: a colon **:** and 4 spaces of indentation.



EXAMPLE

Counting from 1 to 5

Code

```
count = 1
while count <= 5:
    print("Hello", count)
    count = count + 1

print("Done!")
```

Output

```
Hello 1
Hello 2
Hello 3
Hello 4
Hello 5
Done!
```

STEP BY STEP

What Python does each round

count	count <= 5 ?	What happens
1	True	prints Hello 1 → count becomes 2
2	True	prints Hello 2 → count becomes 3
…	…	…
5	True	prints Hello 5 → count becomes 6
6	False	loop stops → prints Done!

REMEMBER

Every loop has 3 parts

1

Start

```
count = 1
```

Create the counter before the loop

2

Condition

```
count <= 5
```

Checked before every round

3

Update

```
count += 1
```

Move towards the end

⚠️ WATCH OUT

Infinite loops

If the condition **never** becomes False, the loop runs **forever**.

Stop it: press ■ Stop in Colab / Jupyter, or Ctrl + C in a terminal.

```
count = 1
while count <= 5:
    print("Hello")
    # forgot count += 1
    # count is ALWAYS 1!
```

LOOPS + INPUT

Repeat until the user says stop

Code

```
word = ""
while word != "quit":
    word = input("Type a word: ")
    print("You typed:", word)

print("Goodbye! 👋")
```

Output

```
Type a word: apple
You typed: apple
Type a word: quit
You typed: quit
Goodbye! 👋
```

We don’t know how many times it will repeat — that’s when **while** is perfect.

INPUT VALIDATION

Keep asking until the answer is valid

Code

```
rating = int(input("Rate 1-5: "))

while rating < 1 or rating > 5:
    print("Please enter 1 to 5!")
    rating = int(input("Rate 1-5: "))

print("Thanks for rating", rating)
```

Output

```
Rate 1-5: 9
Please enter 1 to 5!
Rate 1-5: 0
Please enter 1 to 5!
Rate 1-5: 3
Thanks for rating 3
```

CONTROLLING THE LOOP

break and continue

break — leave now

```
while True:
    pw = input("Password: ")
    if pw == "python123":
        print("Welcome!")
        break
```

Stops the loop immediately.

continue — skip

```
n = 0
while n < 10:
    n += 1
    if n == 5:
        continue
    print(n)
```

Skips this round (prints 1–10 without 5).

🎮 GAME UPGRADE

Guess the Number — now with a loop

```
import random
secret = random.randint(1, 20)
tries = 0
guess = 0
while guess != secret:
    guess = int(input("Your guess: "))
    tries += 1
    if guess < secret:
        print("Too low!")
    elif guess > secret:
        print("Too high!")

print("Correct! Tries:", tries)
```

On Day 3 the player had only **one** guess.

Now the loop keeps going **until the guess is right**.

We also **count the tries** with **tries += 1**.

⭐ Challenge: give the player only **5 tries**.

🧠 QUICK CHECK

Predict before you run!

Q1. How many times?

```
i = 0
while i < 3:
    print("Hi")
    i += 1
```

Q2. What's wrong?

```
x = 10
while x > 0:
    print(x)
```

Q3. What is printed?

```
n = 1
while n < 20:
    n = n * 2
print(n)
```

SUMMARY

What we learned today

Code	Meaning
while count <= 5:	repeat while the condition is True
count += 1	update the counter so the loop can end
infinite loop	condition never False → press ■ Stop
while True:	loop until something happens…
break	…then leave the loop immediately
continue	skip to the next round

💻 YOUR TURN

Practice time!

Open **exercises.ipynb** in the Day 4 folder and solve the 10 tasks, from ⭐ to ⭐ ⭐ ⭐.

⭐ Countdown from 10 to 1

⭐ ⭐ Adding machine until “done”

⭐ ⭐ ⭐ Piggy bank 🐷

Next lesson: for loops and range() 🔁